

Ubuntu Server Hardening Evidence Report

Ubuntu Server Hardening Evidence Report

Generated on: Mon 6 Jul 19:00:53 UTC 2026 Evidence source: /var/log/server-hardening/latest

This report is a human-readable summary generated from hardening evidence files. Review and sanitize before publishing screenshots or excerpts publicly.

1. Hardening Summary

Ubuntu Server Hardening Summary

=====

Timestamp: 20260706-180312
Script version: 2.0.0-final
Hostname: blue-team
Report directory: /var/log/server-hardening/20260706-180312
Latest symlink: /var/log/server-hardening/latest

Options:

- Audit only: false
- SSH port: 22
- SSH port changed: false
- SSH password authentication disabled: false
- Web ports allowed: false
- Admin user configured: none
- Root password locked: true
- IPv6 disabled by script: false
- RKHunter enabled: true

Key evidence:

- Pre-hardening Lynis log: /var/log/server-hardening/20260706-180312/lynis-pre-hardening.log
- Post-hardening Lynis log: /var/log/server-hardening/20260706-180312/lynis-post-hardening.log
- Pre-hardening Lynis report: /var/log/server-hardening/20260706-180312/lynis-report-pre-hardening.pdf
- Post-hardening Lynis report: /var/log/server-hardening/20260706-180312/lynis-report-post-hardening.pdf

- UFW status: /var/log/server-hardening/20260706-180312/ufw-status-final.txt
- Fail2Ban status: /var/log/server-hardening/20260706-180312/fail2ban-sshd-status.txt
- Effective SSH config: /var/log/server-hardening/20260706-180312/sshd-effective-config.txt
- SSH service status: /var/log/server-hardening/20260706-180312/ssh-service-status.txt
- Listening ports before/after: /var/log/server-hardening/20260706-180312/listening-ports-before-after.txt

Useful commands:

- List reports: `sudo find /var/log/server-hardening -maxdepth 2 -type f | sort`
- View latest summary: `sudo cat /var/log/server-hardening/latest/hardening-summary.txt`
- View latest Lynis post-hardening log: `sudo less /var/log/server-hardening/latest/lynis-post-hardening.log`
- Check Lynis hardening index: `sudo grep -i "hardening index" /var/log/server-hardening/latest/lynis-post-hardening.log`
- Check Lynis warnings: `sudo grep -i "warning" /var/log/server-hardening/latest/lynis-post-hardening.log`

Next actions:

1. Review Lynis warnings and suggestions.
2. Confirm application-specific ports and services.
3. Check SSH access from a second terminal before ending the current session.
4. Integrate logs with SIEM tooling such as Wazuh, Microsoft Sentinel, or Splunk.
5. Schedule regular patching, vulnerability scanning, and configuration reviews.

2. Lynis Hardening Index

2026-07-06 18:10:54 Hardening index : [77] [#####]]

3. Lynis Warnings

2026-07-06 18:10:02 Warning: MongoDB instance allows any user to access databases [test:DBS-1000]
 2026-07-06 18:10:42 Skipped test CONT-8104 (Checking Docker info for any warnings)

4. Lynis Suggestions

2026-07-06 18:08:50 Suggestion: Install libpam-tmpdir to set \$TMP and \$TMPDIR for PAM sessions [test:TMP-1000]
 2026-07-06 18:08:50 Suggestion: Install apt-listbugs to display a list of critical bugs prior to upgrading [test:APT-1000]
 2026-07-06 18:08:50 Suggestion: Install apt-listchanges to display any significant changes prior to upgrading [test:APT-1000]
 2026-07-06 18:08:50 Suggestion: Copy /etc/fail2ban/jail.conf to jail.local to prevent it being overwritten [test:FAI-1000]
 2026-07-06 18:08:52 Suggestion: Set a password on GRUB boot loader to prevent altering boot loader [test:GRUB-1000]
 2026-07-06 18:08:55 Suggestion: Determine runlevel and services at startup [test:BOOT-5180]
 2026-07-06 18:09:03 Suggestion: Consider hardening system services [test:BOOT-5264] [details:BOOT-5264]
 2026-07-06 18:09:07 Suggestion: Check PAM configuration, add rounds if applicable and expire dates [test:PAM-1000]
 2026-07-06 18:09:07 Suggestion: Configure password hashing rounds in /etc/login.defs [test:PAM-1000]
 2026-07-06 18:09:10 Suggestion: When possible set expire dates for all password protected accounts [test:PAM-1000]
 2026-07-06 18:09:11 Suggestion: Default umask in /etc/login.defs could not be found and default umask is 002 [test:PAM-1000]
 2026-07-06 18:09:14 Suggestion: To decrease the impact of a full /home file system, place /home on a separate partition [test:PAM-1000]
 2026-07-06 18:09:14 Suggestion: To decrease the impact of a full /var file system, place /var on a separate partition [test:PAM-1000]
 2026-07-06 18:09:18 Suggestion: Disable drivers like USB storage when not used, to prevent unauthorized access [test:PAM-1000]
 2026-07-06 18:09:19 Suggestion: Check DNS configuration for the dns domain name [test:NAME-1000]

```

2026-07-06 18:09:40 Suggestion: Install debsums utility for the verification of packages with
2026-07-06 18:09:47 Suggestion: Install package apt-show-versions for patch management purposes
2026-07-06 18:09:54 Suggestion: Determine if protocol 'dccp' is really needed on this system
2026-07-06 18:09:54 Suggestion: Determine if protocol 'sctp' is really needed on this system
2026-07-06 18:09:54 Suggestion: Determine if protocol 'rds' is really needed on this system
2026-07-06 18:09:54 Suggestion: Determine if protocol 'tipc' is really needed on this system
2026-07-06 18:10:01 Suggestion: Consider hardening SSH configuration [test:SSH-7408] [details:-] [solution:-]
2026-07-06 18:10:10 Suggestion: Enable logging to an external logging host for archiving purposes
2026-07-06 18:10:11 Suggestion: Check what deleted files are still in use and why. [test:LOCAL-1000] [details:-] [solution:-]
2026-07-06 18:10:13 Suggestion: Add a legal banner to /etc/issue, to warn unauthorized users
2026-07-06 18:10:15 Suggestion: Enable process accounting [test:ACCT-9622] [details:-] [solution:-]
2026-07-06 18:10:44 Suggestion: Install a file integrity tool to monitor changes to critical files
2026-07-06 18:10:44 Suggestion: Determine if automation tools are present for system management
2026-07-06 18:10:48 Suggestion: Consider restricting file permissions [test:FILE-7524] [details:-] [solution:-]
2026-07-06 18:10:53 Suggestion: One or more sysctl values differ from the scan profile and could be hardened
2026-07-06 18:10:53 Suggestion: Harden compilers like restricting access to root user only [test:LOCAL-1000] [details:-] [solution:-]

```

5. Firewall Status

```

Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

```

To	Action	From
--	-----	----
22/tcp	ALLOW IN	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)

6. Fail2Ban SSH Jail Status

```

Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed: 0
|   `-- Journal matches: _SYSTEMD_UNIT=ssh.service + _COMM=sshd
`-- Actions
    |- Currently banned: 0
    |- Total banned: 0
    `-- Banned IP list:

```

7. SSH Effective Configuration - Key Controls

```

port 22
maxauthtries 3
maxsessions 2

```

```

clientaliveinterval 300
clientalivecountmax 2
permitrootlogin no
pubkeyauthentication yes
passwordauthentication yes
kbdinteractiveauthentication no
x11forwarding no
tcpkeepalive no
allowtcpforwarding no
allowagentforwarding no
banner /etc/issue.net
loglevel VERBOSE

```

8. SSH Service Status

```

ssh.service - OpenBSD Secure Shell server
    Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
    Active: active (running) since Mon 2026-07-06 18:07:03 UTC; 50ms ago
    Invocation: 1bce92dc48484e5694ab006b66132f0c
    TriggeredBy: ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
    Process: 104724 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 104727 (sshd)
    Tasks: 1 (limit: 1429)
    Memory: 1.3M (peak: 2.2M)
    CPU: 49ms
    CGroup: /system.slice/ssh.service
            104727 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

```

```

Jul 06 18:07:03 blue-team systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jul 06 18:07:03 blue-team sshd[104727]: Server listening on 0.0.0.0 port 22.
Jul 06 18:07:03 blue-team sshd[104727]: Server listening on :: port 22.
Jul 06 18:07:03 blue-team systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

```

9. Listening Ports After Hardening

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*	users:(("systemd-reso
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*	users:(("systemd-reso
udp	UNCONN	0	0	192.168.21.41%ens33:68	0.0.0.0:*	users:(("systemd-netw
udp	UNCONN	0	0	127.0.0.1:323	0.0.0.0:*	users:(("chronyd",pi
udp	UNCONN	0	0	:::1:323	::::*	users:(("chronyd",pi
tcp	LISTEN	0	4096	127.0.0.1:27017	0.0.0.0:*	users:(("mongod",pid-
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*	users:(("systemd-reso
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*	users:(("systemd-reso

tcp	LISTEN	0	511	0.0.0.0:3000	0.0.0.0:*	users:(("node",pid=50
tcp	LISTEN	0	4096	0.0.0.0:22	0.0.0.0:*	users:(("sshd",pid=10
tcp	LISTEN	0	511	*:43005	*:*	users:(("node",pid=50
tcp	LISTEN	0	4096	*:9090	*:*	users:(("prometheus",
tcp	LISTEN	0	4096	:::22	:::*	users:(("sshd",pid=10

10. Enabled Services After Hardening - First 100 Lines

UNIT FILE	STATE	PRESET
snap-core20-2866.mount	enabled	enabled
snap-core24-1643.mount	enabled	enabled
snap-prometheus-86.mount	enabled	enabled
snap-rocketchat\x2dserver-1788.mount	enabled	enabled
snap-snapd-27406.mount	enabled	enabled
apport-autoreport.path	enabled	enabled
tpm-udev.path	enabled	enabled
apparmor.service	enabled	enabled
apport.service	enabled	enabled
audit-rules.service	enabled	enabled
auditd.service	enabled	enabled
blk-availability.service	enabled	enabled
chrony.service	enabled	enabled
cloud-config.service	enabled	enabled
cloud-final.service	enabled	enabled
cloud-init-local.service	enabled	enabled
cloud-init-main.service	enabled	enabled
cloud-init-network.service	enabled	enabled
console-setup.service	enabled	enabled
cron.service	enabled	enabled
dmesg.service	enabled	enabled
e2scrub_reap.service	enabled	enabled
fail2ban.service	enabled	enabled
finalrd.service	enabled	enabled
getty@.service	enabled	enabled
gpu-manager.service	enabled	enabled
grub-initrd-fallback.service	enabled	enabled
grub2-common.service	enabled	enabled
kdump-tools.service	enabled	enabled
keyboard-setup.service	enabled	enabled
lvm2-monitor.service	enabled	enabled
ModemManager.service	enabled	enabled
multipathd.service	enabled	enabled
netplan-configure.service	enabled	enabled
networkd-dispatcher.service	enabled	enabled
open-iscsi.service	enabled	enabled
open-vm-tools.service	enabled	enabled

pollinate.service	enabled	enabled
rsyslog.service	enabled	enabled
secureboot-db.service	enabled	enabled
setvtrgb.service	enabled	enabled
snap.prometheus.prometheus.service	enabled	enabled
snap.rocketchat-server.rocketchat-mongo.service	enabled	enabled
snap.rocketchat-server.rocketchat-server.service	enabled	enabled
snapd.apparmor.service	enabled	enabled
snapd.autoimport.service	enabled	enabled
snapd.core-fixup.service	enabled	enabled
snapd.recovery-chooser-trigger.service	enabled	enabled
snapd.seeded.service	enabled	enabled
snapd.service	enabled	enabled
snapd.system-shutdown.service	enabled	enabled
sshd-keygen.service	enabled	enabled
sysstat.service	enabled	enabled
systemd-networkd-wait-online.service	enabled	enabled
systemd-networkd.service	enabled	enabled
systemd-pstore.service	enabled	enabled
systemd-resolved.service	enabled	enabled
thermald.service	enabled	enabled
ua-reboot-cmds.service	enabled	enabled
ubuntu-advantage.service	enabled	enabled
udisks2.service	enabled	enabled
ufw.service	enabled	enabled
unattended-upgrades.service	enabled	enabled
vgauth.service	enabled	enabled
system-xfs_scrub.slice	enabled	enabled
apport-forward.socket	enabled	enabled
cloud-init-hotplugd.socket	enabled	enabled
dm-event.socket	enabled	enabled
iscsid.socket	enabled	enabled
lvm2-lvmpolld.socket	enabled	enabled
lxd-installer.socket	enabled	enabled
polkit-agent-helper.socket	enabled	enabled
snapd.socket	enabled	enabled
ssh.socket	enabled	enabled
systemd-networkd.socket	enabled	enabled
systemd-resolved-monitor.socket	enabled	enabled
systemd-resolved-varlink.socket	enabled	enabled
uidd.socket	enabled	enabled
remote-fs.target	enabled	enabled
apport-autoreport.timer	enabled	enabled
apt-daily-upgrade.timer	enabled	enabled
apt-daily.timer	enabled	enabled
dpkg-db-backup.timer	enabled	enabled

e2scrub_all.timer	enabled	enabled
fstrim.timer	enabled	enabled
fwupd-refresh.timer	enabled	enabled
logrotate.timer	enabled	enabled
lynis.timer	enabled	enabled
man-db.timer	enabled	enabled
mdcheck_continue.timer	enabled	enabled
mdcheck_start.timer	enabled	enabled
mdmonitor-oneshot.timer	enabled	enabled
motd-news.timer	enabled	enabled
snapped.snap-repair.timer	enabled	enabled
sysstat-collect.timer	enabled	enabled
sysstat-rotate.timer	enabled	enabled
sysstat-summary.timer	enabled	enabled
ua-timer.timer	enabled	enabled
update-notifier-download.timer	enabled	enabled

11. Auditd Status

auditd.service - Security Audit Logging Service

Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: enabled)

Active: active (running) since Mon 2026-07-06 18:07:09 UTC; 43s ago

Invocation: 4656956136de41b1b9fe9730af8dbd67

Docs: man:auditd(8)

<https://github.com/linux-audit/audit-documentation>

Process: 105374 ExecStart=/usr/sbin/auditd (code=exited, status=0/SUCCESS)

Main PID: 105375 (auditd)

Tasks: 2 (limit: 1429)

Memory: 836K (peak: 2M)

CPU: 49ms

CGroup: /system.slice/auditd.service

105375 /usr/sbin/auditd

Jul 06 18:07:09 blue-team systemd[1]: Starting auditd.service - Security Audit Logging Service

Jul 06 18:07:09 blue-team auditd[105375]: No plugins found, not dispatching events

Jul 06 18:07:09 blue-team auditd[105375]: Init complete, auditd 4.1.2 listening for events

Jul 06 18:07:09 blue-team systemd[1]: Started auditd.service - Security Audit Logging Service

12. Follow-up Actions

- Review remaining Lynis warnings and suggestions.
- Confirm application-specific ports and services.
- Test SSH key access from a second terminal before disabling SSH password authentication.
- Integrate logs with Wazuh, Microsoft Sentinel, or Splunk.

- Schedule regular patching, vulnerability scanning and configuration reviews.